

Rishman Jot Kaur Chahal

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Education

- **PhD**, IIT Kanpur, Kanpur, India 2020
- **M.Sc.** General Economics, Madras School of Economics (MSE), Chennai, India 2015
- **B.A.(H)** Economics, Delhi University (DU), New Delhi, India 2013

Affiliations

- **Assistant Professor of Economics**, Dept. of HSS, IIT Roorkee Nov 2020 – contd.
- **Joint Faculty**, Mehta Family School of DS and AI, IIT Roorkee June 2021 – contd.
- **Visiting Faculty**, Gati Shakti Vishwavidyalaya, Vadodara, India July 2023 - Nov 2023
- **Research Intern**, Secretariat of the Convention of Biodiversity, UNEP, Montreal, Canada Sept 2019 – Mar 2020

Professional Recognition/ Awards

- **Best Paper Award** (2026) at the Symposium on Ecology, Governance & Technology, organised by Mehta Family School of Sustainability, IIT Indore, India
- **Best Paper Award** (2025), International Conference on Innovations in Finance, Inclusion & Economic Growth, IIT Dhanbad, India
- **Third Position in Datathon Challenge** (2023) by National Institute of Urban Affairs (NIUA), Ministry of Housing and Urban Affairs (MoHUA)
- **Travel Grant** (2018) as Student Research Award by Dean of Resources and Alumni, IIT Kanpur
- **Passed National Eligibility Test (NET) in Economics** (2014) conducted by UGC
- **Awarded Merit Certificate in Economics** (given only to top 0.1% successful candidates in HSSC) (2010) by CBSE

Research Interest

Corporate Governance and Institutions, Financial Economics, Sustainable Finance, Computational General Equilibrium (CGE) Modelling, Time Series and Machine learning applications in Macroeconomics.

Sponsored Research Projects

- “Networks and Sentiment Sensitivity of Firms”, by **IIT Roorkee**. (Rs. 20L) (20 Jan 2022 - 31 Jan 2024)
- “Developing Deep Learning-based Strategies, Tools and Apps in Global Financial Markets”, by **SmartAlpha Pvt. Ltd.** (Rs. 33.18L) (09 May 2022 - 28 Feb 2025)
- “Embedding Crowding to Passenger Information System using Electronic Ticketing Data”, by **TIH-IIT Kharagpur** (Rs. 31.68L) (18 Sept 2023 - 12 Sept 2025)
- As a Co-PI, “Advanced Transport Modeling and Simulation for National Highways in India” by **IAHE, MoRTH** (Rs.400L) (25 Mar 2021-Present)
- “Char Dham Ropeway Project: Assessment on Lives and Livelihood”, by **ICSSR** (Rs. 14.30 lacs) 06 Apr 2026 - Present
- As a Co-PI, “CO-INVENT: COoperative project between India and Norway on studying cryo-climatology of the glaciated Himalayan region” by **UGC**, India (NOK 27,99,207) (03 Nov 2025-Present)

Training

- Faculty instructor for eMasters course on Machine Learning for Climate and Carbon Finance by IIT Kanpur (Spring 2025-26)
- Modules on Machine learning and Time Series modeling to train executives from Deloitte Inc. (Batch 1-8, 2022-25).
- Involved in training program for professionals from Income Tax Department to teach Data Analytics and time series modelling using Business Intelligence Tools (Batch 1-4, August 2024).
- Visiting Faculty at Gati Shakti Vishwavidyalaya (GSV), Vadodara, to teach Engineering Economics (Autumn 2023).

- Faculty instructor for eMasters course on Derivatives Market and Risk Management by IIT Kanpur (2022)
- Distinguished Resource Person for a workshop entitled, “Financial Time-Series Modelling in R” (FinMODE), IIT Kanpur (2018, 2021, 2022)
- Faculty instructor for eMasters course on Quantitative Finance and Risk Management by IIT Kanpur (2021)

Journal Publications

- **Chahal, Rishman Jot Kaur**, Bidasaria, Hemant, Khan, Hera Asif, and Ahmad, Wasim, “Do global bond market sentiments transmit to green bonds? Evidence from a quantile connectedness framework”, *Journal of Behavioral and Experimental Finance*, 2026, DOI: <https://doi.org/10.1016/j.jbef.2026.101151>
- Manish, and **Chahal, Rishman Jot Kaur**, “Bridging behavioral insights and quantitative finance: AI-powered Black–Litterman framework with technical and sentiment signals”, *Research in International Business and Finance*, 2026, DOI: <https://doi.org/10.1016/j.ribaf.2026.103329>
- Khan, Hera Asif, and **Chahal, Rishman Jot Kaur**, “The evolving influence of news and social media sentiments on Indian sectoral returns during COVID-19: TVP-VAR extended joint connectedness approach”, *Review of Behavioral Finance*, 2025, DOI: <https://doi.org/10.1108/RBF-04-2025-0140>.
- Khan, Hera Asif, and **Chahal, Rishman Jot Kaur**, “Asymmetric impact of social media sentiments and stock market uncertainty on Indian sectoral returns: A quantile-on-quantile approach”, *The North American Journal of Economics and Finance*, 2025, DOI: <https://doi.org/10.1016/j.najef.2025.102456>.
- **Chahal, Rishman Jot Kaur**, and Ahmad, Wasim, “Political Connections and Corporate Debt Structure: Moderating effect of Government-owned Banks of India”, *Abacus*, 2025, DOI: <https://doi.org/10.1111/abac.70002>.
- Ahmad, Wasim, **Chahal, Rishman Jot Kaur**, and Rais, Sheren, “Understanding the impact of the coronavirus outbreak on the economic integration of ASEAN countries”, *Asia and the Global Economy*, 2022, DOI: <https://doi.org/10.1016/j.aglobe.2022.100040>
- Ahmad, Wasim, Kutan, Ali M., **Chahal, Rishman Jot Kaur**, and Kattumuri, Ruth, “COVID-19 Pandemic and firm-level dynamics in the USA, UK, Europe, and Japan”, *International Review of Financial Analysis*, 2021, DOI: <https://doi.org/10.1016/j.irfa.2021.101888>
- Ahmad, Wasim, Prakash, Ravi, Uddin, Gazi Salah, **Chahal, Rishman Jot Kaur**, Rahman, Md. Lutfur, and Dutta, Anupam, “On the Intraday Dynamics of Oil Price and Exchange Rate: What can we learn from China and India?”, *Energy Economics*, 2020, DOI: <https://doi.org/10.1016/j.eneco.2020.104871>
- **Chahal, Rishman Jot Kaur**, and Ahmad, Wasim, “Political Connections, Investment Inefficiency and the Indian Banking Crisis”, *Quarterly Review of Economics and Finance*, 2020, DOI: <https://doi.org/10.1016/j.qref.2020.10.016>

ADBI Working Papers and Book Chapters

- Sahu, Parthasarathi, and **Chahal, Rishman Jot Kaur**, “Climate Change and Fiscal Policy in South Asia: A Symmetric and Asymmetric Analysis of Urbanization Effects in India” (2026) accepted for publication as a book chapter for an ADBI-led book.
- Gupta, Garima, Gulati, Rachita, and **Chahal, Rishman Jot Kaur**, “Central bank digital currency and multidimensional bank stability index: Does monetary policy play a moderating role?” (2025), *ADBI Working Paper Series*, 2025, No. 1509, DOI: <https://doi.org/10.56506/ACZW4884>. [Also accepted as a book chapter in the book: “Emerging Challenges in Banking in Asia and the Pacific”, published by SPRINGER NATURE.]
- **Chahal, Rishman Jot Kaur**, and Ahmad, Wasim, “Linking Corporate and Environmental Governance to the Sustainable Development Goals in an Emerging Economy” (2024), In: *Striving to Meet the Sustainable Development Goals: Next Steps for Policymakers and Practitioners*, ADBI, ISBN: 978-4-89974-324-8. DOI: <https://doi.org/10.56506/ACFH4829>
- Ahmad, Wasim, **Chahal, Rishman Jot Kaur**, “Do Political Connections Spur Technology Transfer? Evidence from Governance Perspective in India”, *ADBI Working Paper Series*, 2023, No. 1362.

Other Book Chapters

- Sehgal, Sanjay, Ahmad, Wasim and **Chahal, Rishman Jot Kaur**, “On the determinants of Economic Integration in SAARC countries: A Macroeconomic Perspective” (2020), In: *Economic and Financial Integration in South Asia*, Taylor & Francis, UK. ISBN: 9780815380139.

Work in Progress (relevant)

- Ahmad, Danish, **Chahal, Rishman Jot Kaur**, “Investment efficiency and the mediating role of CSR: Evidence from the altered Board network position of Indian firms”, *Quarterly Review of Economics and Finance*, 2026 (revision submitted)
- Ahmad, Danish, **Chahal, Rishman Jot Kaur**, “When Politics Meet Accounting: The Moderating Effect of Corporate Political Connections on the Reporting Quality-Investment Efficiency Nexus in India”, *Journal of Financial Reporting and Accounting*, 2026 (revision submitted)
- Khan, Hera Asif, and **Chahal, Rishman Jot Kaur**, “Do uncertainties and investor sentiment affect traditional and sustainable assets differently across quantiles and frequency horizons?”, *Financial Research Letters*, 2026 (revision submitted)
- Yadav, Kavita, **Chahal, Rishman Jot Kaur**, “Reconstructing the Digital Financial Inclusion Index using the Benefit-of-the-Doubt (BoD) Approach: Implications for sustainable growth”, *International Journal of Development Issues*, 2026 (revision submitted)
- Yadav, Kavita, **Chahal, Rishman Jot Kaur**, “Digital Payment Index and Bank Stability in Emerging Economies: Evidence from India”, *Eurasian Economic Review*, 2026 (revision submitted)
- Manish, **Chahal, Rishman Jot Kaur**, “Portfolio optimization under alternative risk measures: Sentiment-informed AI forecasts and strategic commodity exposure” (under revision)
- **Chahal, Rishman Jot Kaur**, Ahmand, Danish, Ahmad, Wasim, “Political Connections and Environmental Compliance in a Federal Economy: Evidence from India” (under review)
- Gupta, Garima, **Chahal, Rishman Jot Kaur**, Gulati, Rachita, “Unveiling monetary policy evolution and future research horizons through bibliometric and content analysis” (under review)
- Gupta, Garima, Khan, Hera Asif, **Chahal, Rishman Jot Kaur**, “CBDC attention and uncertainty spillovers across asset returns and market sentiments: Implications from an emerging financial market” (under review)
- Manish, **Chahal, Rishman Jot Kaur**, “Uncovering energy transition signals: Mapping multi-channel drivers of rare earth equity markets with explainable deep learning”, 2026 (under review)
- Manish, **Chahal, Rishman Jot Kaur**, “Predictive drivers of a Paris-aligned ESG index: An explainable deep learning approach”, 2026 (under review)

Workshops & Conference Presentations/Publications

- Fedujwar, Rupam, Agarwal, Amit, **Chahal, Rishman Jot Kaur** (2026). “To board or To wait? Investigating how real time crowding information shapes passengers boarding behaviour”, In: The 15th Asia-Pacific Conference on Transportation and the Environment, Jeju Island, South Korea.
- Yadav, Yogesh, **Chahal, Rishman Jot Kaur**, Agarwal, Amit. “ML-based Prediction of District-Level GDP and Mapping of Inequalities in GDP Distribution in India”. In: Symposium on Ecology, Governance and Technology, IIT Indore, Indore, India. (2026) [**Best Paper Award**]
- Pallath, Sneha, **Chahal, Rishman Jot Kaur**. “Understanding the Tone, Uncertainty and Emotion in RBI’s CBDC Communication: A Textual Analysis Approach”. In: Global Finance Conclave 2026. O.P Jindal Global University, Sonapat, India. (2026)
- Sahu, Parthasarathi, **Chahal, Rishman Jot Kaur**. “Symmetric and Asymmetric Impacts of Climate Shocks on Fiscal Policy: Urbanization as a Transmitting Mechanism in India”. In: ADBI-SAU Conference on Climate Change and Resilience in South Asia, Asian Development Bank Institute (ADBI) and South Asian University (SAU). (2025)
- Sahu, Parthasarathi, **Chahal, Rishman Jot Kaur**. “Unpacking the Urban-Climate Puzzle: Lessons from Six Decades of Indian Development”. In: 13th World Congress of the Econometric Society (ESWC 2025). Organized by Korean Economic Association, Seoul National University, Korea University, Sogang University, Sungkyunkwan University, Yonsei University, Bank of Korea, Korea Development Institute (KDI), and Korea Institute for International Economic Policy (KIEP), Seoul, South Korea. (2025)
- Sahu, Parthasarathi, **Chahal, Rishman Jot Kaur**. “Climate-Urban Nexus in India: A Time Series Analysis of Economic Growth and Environmental Interactions”. In: International Conference on Enabling Sustained Growth in Emerging Asia (ESGEA). University of Hyderabad, Hyderabad, India. (2025)
- Sahu, Parthasarathi, **Chahal, Rishman Jot Kaur**. “Dynamics of Climate Change and Urbanization in India: A Time Series Analysis Using VECM Approach”. In: 59th Annual Conference of the The Indian Econometric Society (TIES). Banaras Hindu University, Varanasi, India. (2025)
- Yadav, Kavita, **Chahal, Rishman Jot Kaur**. “Reconstructing the Digital Financial Inclusion Index through Benefit-of-the-Doubt (BoD) Approach: Further determining growth and policy implications”. In: Empowering Viksit

Bharat@2047: Innovations in Finance, Inclusion, & Economic Growth, IIT (ISM), Dhanbad. India. (2025) [**Best Paper Award**]

- Yadav, Kavita, **Chahal, Rishman Jot Kaur**. "Revisiting the Dimensions of Digital Financial Inclusion: A Benefit-of-the-Doubt (BoD) Approach". In: The 59th Annual Conference of the Indian Econometric Society (TIES), Banaras Hindu University (BHU), Varanasi, India. (2025)
- Ahmad, Danish, Srivasatava, Komal, **Chahal, Rishman Jot Kaur**. "When Politics Meet Accounting: The Moderating effect of Corporate Political Connections on the Reporting Quality-Investment Efficiency Nexus in India". In: The 59th Annual Conference of the Indian Econometric Society (TIES), Banaras Hindu University (BHU), Varanasi, India. (2025)
- Gupta, Garima, Gulati, Rachita, **Chahal, Rishman Jot Kaur**. "Central bank digital currency and multidimensional bank stability index: Does monetary policy play a moderating role?". In: From Crisis to Resilience Shaping the Future of Banking in Asia Pacific. Organized by Asian Development Bank Institute (ADBI), Japan and South Asian University, India. (2025)
- Gupta, Garima, Khan, Hera Asif, **Chahal, Rishman Jot Kaur**. "CBDC attention and uncertainty spillovers across asset returns and market sentiments: Implications from an emerging financial market". In: 1st International Conference on Sustainability and Economic Growth amid Global Uncertainty. Organized by The Applied Economics Society (TAES) and National Institute of Technology, Goa. (2025)
- Gupta, Garima, Khan, Hera Asif, **Chahal, Rishman Jot Kaur**. "CBDC attention and uncertainty spillovers across asset returns and market sentiments: Implications from an emerging financial market". In: International Conference in Banking and Financial Studies (ICBFS 2025). Organized by The Catholic University of the Sacred Heart, Piacenza-Milan, Italy. (2025)
- Gupta, Garima, Gulati, Rachita, **Chahal, Rishman Jot Kaur**. "Monetary Policy: Evolution, Transmission Dynamics, and Future Horizons through Bibliometric Analysis". In: Bibliometrics and Research Impact Community (BRIC) Conference. Organized by University of Montreal, Canada. (2025)
- Manish, **Chahal, Rishman Jot Kaur**. "Hedging volatility with intelligence: AI-enhanced portfolio optimization using alternative risk measures and commodity futures". In: International Conference on Financial Markets and Corporate Finance, VGSOM IIT Kharagpur. (2025)
- Manish, **Chahal, Rishman Jot Kaur**. "Integrating technical and sentiment indicators in the Black-Litterman model: Deep learning insights in different market conditions". In: International Conference on Financial Markets and Corporate Finance, VGSOM, IIT Kharagpur. (2025)
- Fedujwar, Rupam, Rawat, Nishtha, Agarwal, Amit, **Chahal, Rishman Jot Kaur**. "From variability to uniformity: Standardization of electronic ticketing machines data for urban transit systems in India". In: 8th Conference of Transportation Research Group of India (CTRG), Guwahati, India. (2025).
- Fedujwar, Rupam, Agarwal, Amit and **Chahal, Rishman Jot Kaur** (2025). "Quantification of crowding discomfort in public transport: Results from an Indian mid-size city". In: 8th Conference of Transportation Research Group of India (CTRG), Guwahati, India.
- Rawat, Nishtha, Jeengar, Kuldeep, Agarwal, Amit and **Chahal, Rishman Jot Kaur**. (2024). "Boarding Alighting Counting in Different Transit Vehicles under Crowded Conditions". In IEEE CONECCT 2024 at IISc Bangalore. DOI:10.1109/CONECCT62155.2024.10677074. (2024)
- **Chahal, Rishman Jot Kaur**, Bidasaria Hemant, Khan, Hera Asif and Ahmad, Wasim (2024), "Quantile connectedness between conventional bond market sentiments and Global Green Bond Returns: Implications for effective hedging". In: 6th PAN-IIT International Conference of Financial Markets & Corporate Finance, IIT(ISM) Dhanbad, India. (2024)
- Khan, Hera Asif and **Chahal, Rishman Jot Kaur**, "Asymmetric impact of social media sentiments and stock market uncertainty on Indian sectoral returns: A quantile-on-quantile approach" presented at the 14th Financial Markets & Corporate Governance Conference (FMCG), Monash University, Malaysia. (2024)
- Khan, Hera Asif and **Chahal, Rishman Jot Kaur**, "Asymmetric impact of social media sentiments and stock market uncertainty on Indian sectoral returns: A quantile-on-quantile approach" In: the 99th Western Economic Association International (WEAI), Seattle, USA. (2024)
- **Chahal, Rishman Jot Kaur**, Bidasaria, Hemant, Khan, Hera Asif and Ahmad, Wasim (2024), "Quantile connectedness between conventional bond market sentiments and Global Green Bond Returns: Implications for effective hedging" presented at the 11th International Conference on Finance and Banking (FIBA), Bucharest University of Economic Studies, Romania. (2024)
- Sahu, Parthasarathi and **Chahal, Rishman Jot Kaur**. "General Equilibrium Approach to Evaluate Economic Growth and Welfare Policy Analysis: Network and Meta-Analysis in a Systematic Review". In: 64th Annual Conference of

- the Indian Society of Labour Economics (ISLE). Organized by University of Hyderabad, Hyderabad, India. (2024)
- Gupta, Garima, Gulati, Rachita, **Chahal, Rishman Jot Kaur**. "Central bank digital currency and multidimensional bank stability index: Does monetary policy play a moderating role? Evidence from BRICS". In: DEA 2024: International Conference of Data Envelopment Analysis. Organized by BIMTECH, Noida and University of Surrey, United Kingdom. (2024)
 - Kurray, Suraj Pal, Kumar, Rajat, **Chahal, Rishman Jot Kaur** and Agarwal, Amit. "Forecasting the future of EV and ICEV markets and their associated CO2 emissions in Delhi using the deep learning method. In: India Just Transition Summit 2024. New Delhi, India". (2024)
 - Sain, Jitendra Kumar, Kumar, Rajat, **Chahal, Rishman Jot Kaur** and Agarwal, Amit. "Mapping Kerala's economic landscape: a non-survey approach to constructing regional Input-Output tables for 2018-19". In: 8th International Conference on Economic Structures, Osaka, Japan. 2023
 - Kumar, Rajat, Sahu, Parthasarathi, Agarwal, Amit and **Chahal, Rishman Jot Kaur**. "Wider economic impact at electric vehicle penetration in India: a CGE analysis". In: 51st European Transport Conference. Association for European Transport. Milan, Italy. (2023)
 - Khan, Hera Asif and **Chahal, Rishman Jot Kaur**. "Investor sentiment and stock price crash risk in India" In: International Conference on Applied Economic and Finance (ICAEF), BITS Pilani, India. (2023)
 - Ahmad, Danish and **Chahal, Rishman Jot Kaur**. "Board network position investment efficiency: The mediating influence of ESG on Indian firms". In: Asian Meeting of Econometric Society (AMES-CSW), IIT Delhi, India. (2023)
 - Ahmad, Danish and **Chahal, Rishman Jot Kaur**. "Board network position investment efficiency: The mediating influence of ESG on Indian firms". In: 12th India Finance Conference (IFC), N.L. Dalmia Institute of Management Studies and Research, Mumbai, India. (2023)
 - Ahmad, Danish and **Chahal, Rishman Jot Kaur**. "Board network position investment efficiency: The mediating influence of ESG on Indian firms". In: International Conference on Applied Economic and Finance (ICAEF), BITS Pilani, India. (2023)
 - **Chahal, Rishman Jot Kaur**, Ahmad, Wasim and Bhanumurthy, N. R. "The Trinity Effect of Corporate-Political-Banking relationship over Debt Structure: Evidence from India". Allied Social Sciences Association Meeting (ASSA), American Economic Association (AEA), Chicago, USA. (2021) [A+ Conference, accepted for poster presentation]
 - **Chahal, Rishman Jot Kaur**, Ahmad, Wasim, Uddin, Gazi Salah and Boustanifar, Hamid (2018). "Firm-Political Connections: A Resource Based View Analysis". In: 14th Annual Conference on Economic Growth Development, Indian Statistical Institute Delhi, Delhi, India.
 - Chahal, Rishman Jot Kaur, Ahmad, Wasim (2018). "Overinvestment, Political Connections and Government-Owned Banks: Evidence from India". In: 93rd Western Economic Association International (WEAI), Vancouver, Canada.
 - **Chahal, Rishman Jot Kaur**, Ahmad, Wasim (2018). "Overinvestment, Political Connections and Government-Owned Banks: Evidence from India". In: 3rd International Conference on Financial Markets Corporate Finance (ICFMCF), Indian Institute of Technology Kanpur, Kanpur, India.
 - **Chahal, Rishman Jot Kaur**, Ahmad, Wasim (2017). Politicking Behaviour of Indian Firms. In: Paris Finance Management Conference, Paris, France.
 - **Chahal, Rishman Jot Kaur**, Ahmad, Wasim (2017). "Political Connections and Firm's Performance: Evidence from India". In: 7th Indian Finance Conference (IFC), Indian Institute of Management Bangalore, Bengaluru, India.
 - **Chahal, Rishman Jot Kaur**, Ahmad, Wasim (2017). Politicking Behaviour of Indian Firms. In: 8th Emerging Markets Finance Conference (EMFC), Indira Gandhi Institute of Development Research, Mumbai, India.

Positions of Responsibility

• OC Lab in the HSS Department, IITR	04 June 2025- Jan 2026
• M.Tech DS and M.Tech AI faculty in-charge coordinator (FIC), IITR	July 2024-Jan 2026
• MSc. Seminar and Dissertation Coordinator at the HSS Department, IITR	23 Jul 2024 - June 2025
• Member of SFSC, Mehta Family School of DS & AI, IIT Roorkee	25 Jul 2022 - Dec 2026
• Exam Coordinator at the Mehta Family School of DS and AI, IITR	July 2025-Jan 2026
• Chief Warden of Kasturba Bhawan, Hostel at IITR	July 2022-June 2024
• Member Secretary at the Mehta Family School of DS & AI, IITR	03 Nov 2023 - 23 Jul 2024
• Exam Coordinator at the HSS Department, IITR	July 2022-September 2024
• Warden of Kasturba Bhawan, Hostel at IITR	July 2021-June 2022

International Collaborations

- **UNSW, Sydney** through my collaborative project “Advanced Transport Modeling and Simulation for National Highways in India under CATTS” in collaboration with UNSW, Sydney, Australia, 2022, Agarwal, Amit (IITR, PI); Chahal, R. J.K. (Co-PI) along with others from IITR.
- **University of Bergen, Norway** through my collaborative project “CO-INVENT: COoperative project between India and Norway on studying cryo-climatology of the glaciated Himalayan region”, in collaboration with University of Bergen, Norway, and Jawaharlal Nehru University (JNU), funded by HK-dir and UGC, 2024, Arya, D.S. (IITR, PI) and Chahal, R.J.K. (Co-PI).

Courses Offered

Indian Institute of Technology, Roorkee, 2020–Present

UG Courses: DAI - 101: Introduction to Data Science HSS-01: Macroeconomics, HSN-206 (now HSC-205): Introduction to Econometric Theory, HSN- 305: Advance Topics in Econometric Theory, HSN-352: Building Economics; **PG Courses:** HSN-509 (now HSC-505): Basic Econometrics, HSN-506 Money, Banking and Financial Markets, AID-555 Time Series Data Analysis, AID-562 AI for Investment, HSN - 507: Time Series Analysis and Applications; HS-504 Econometric Lab; **Prepared Elective for PG: HSL – 660 Quantitative Methods for Sustainable Finance**

Technical Strengths

Statistical	STATA (advance), R, Python, EViews
Other Softwares & Tools	C, MS Office, Latex & Beamer (TeX typesetting Program)
Relevant Courses	Applied Econometrics, Time Series Modelling, Machine Learning

Doctoral Candidates

1. Hera Asif Khan Investor sentiment and uncertainty dynamics in securities markets. <i>Department of Humanities and Social Sciences, IIT Roorkee</i> Defence Viva-Voce scheduled on July 02, 2026	2021–2026
2. Parthasarathi Sahu Essays on Urban Agglomeration <i>Department of Humanities and Social Sciences, IIT Roorkee</i>	2021–2026
3. Kavita Yadav Digital Payment Systems and Fintech in Indian Banks, India <i>Department of Humanities and Social Sciences, IIT Roorkee</i> <i>Co-Supervisor: Prof. Rachita Gulati, HSS (IITR) & ADBI, Tokyo</i>	2022–2027
4. Danish Ahmad Essays of Board Networks of ASEAN Economies with a special attention to India <i>Department of Humanities and Social Sciences, IIT Roorkee</i>	2022–2027
5. Garima Gupta Essays on Monetary Policies in Emerging Market Economies <i>Department of Humanities and Social Sciences, IIT Roorkee</i> <i>Co-Supervisor: Prof. Rachita Gulati, HSS (IITR) & ADBI, Tokyo</i>	2022–2027
6. Rajat Kumar Essays in Energy Transition: A CGE Analysis <i>Department of Humanities and Social Sciences, IIT Roorkee</i> <i>Co-Supervisor: Prof. Amit Agarwal, Civil Engineering (IITR)</i>	2023–2028
7. Manish Essays on AI-Driven Portfolio Optimization and Forecasting <i>Mehta Family School of Data Science and Artificial Intelligence, IIT Roorkee</i>	2023–2028
8. Yogesh Yadav Essays on Urban Sprawl in India <i>Department of Humanities and Social Sciences, IIT Roorkee</i>	2024–2029
9. Sneha Pallath Behavioural Narratives of Digital Money	2025–2030

Department of Humanities and Social Sciences, IIT Roorkee

10. Anushka 2025–2030

Climate Transition Risk and Macroeconomic Dynamics: An India-Specific E-DSGE Analysis

Department of Humanities and Social Sciences, IIT Roorkee

11. Suraj Pal Kurrey 2026–2031

A Sentiment-Augmented, Interpretable AI Framework for intelligent credit systems (tentative)

Mehta Family School of Data Science and Artificial Intelligence, IIT Roorkee

12. Mayur Pandey (Part-time) 2026–Present

Algorithmic Governance in Finance: A Universal Constrained Optimization Framework for Safe and Responsible AI

Mehta Family School of Data Science and Artificial Intelligence, IIT Roorkee

Masters Theses Supervised

M.Sc. (Economics) Dissertation

1. Komal Srivastava 2022

Financial Reporting Quality, Political Connection and Investment Efficiency

Department of Humanities and Social Sciences, IIT Roorkee

2. Anupama Vijay 2023

Board Interlocks and performance of the Indian Stock Exchange during COVID-19 crisis

Department of Humanities and Social Sciences, IIT Roorkee

3. Gary Haobam 2023

The Impact of Corporate Governance on Firms' Financial Performance

Department of Humanities and Social Sciences, IIT Roorkee

4. Ahana Saha 2024

Board Interlocks, Deep-in-Debt Firms and Indian Banking Crisis

Department of Humanities and Social Sciences, IIT Roorkee

5. Neha Meena 2025

Research Evolution and Prospects of the Applications of Centrality Measures in Corporate Governance and Organizational Outcomes: A PRISMA-Informed Bibliometric Analysis

Department of Humanities and Social Sciences, IIT Roorkee

6. Alok Varun 2026

Do ETFs Influence Stock Market Volatility? Insights from Indian Markets

Department of Humanities and Social Sciences, IIT Roorkee

7. Arghya Debnath 2026

Sustainable Development of Indian Banks in the Era of Digital Finance

Department of Humanities and Social Sciences, IIT Roorkee

BS-MS (Economics) Batch MS Dissertation

1. Hemant Bidasaria 2026

Climate stress testing for Indian banks

Department of Humanities and Social Sciences, IIT Roorkee

2. Akshat Trehan 2026

Urban Expansion and its Influence on Local Climate: Evidence from Dehradun and Jammu

Department of Humanities and Social Sciences, IIT Roorkee

3. Moulik Gupta 2026

Experimental Validation of Fraud Using Generation of Synthetic Data Outperforming Traditional Methods

Department of Humanities and Social Sciences, IIT Roorkee

M.Tech. Dissertation

1. B Bhanu Prasanth 2021–2023

Portfolio Management based on Technical Indicators using Deep Learning Techniques

M.Tech. (Artificial Intelligence), Mehta Family School of DS & AI, IIT Roorkee

2. Francis Narzary 2022–2024

Estimating District-Level GDP from Night-Time Light Data Using Deep Learning Techniques

M.Tech. (Artificial Intelligence), Mehta Family School of DS & AI, IIT Roorkee

3. Suraj Pal Kurrey 2023–2025
Forecasting the Future of Internal Combustion Engine Vehicles and Electric Vehicles Ownership along with their CO₂ Emission in Delhi
M.Tech. (Artificial Intelligence), Mehta Family School of DS & AI, IIT Roorkee
Co-Supervisor: Prof. Amit Agarwal (Mehta Family School of DS & AI)

4. Abhishek Patel 2023–2025
Decoding Asian Stock Market Predictions: An XAI Inquiry into the Efficacy of Sentiment vs. Technical Indicators
M.Tech. (Artificial Intelligence), Mehta Family School of DS & AI, IIT Roorkee

5. Pranay Majumder 2023–2025
Predicting District-wise GDP based on NTL Data from VIIRS Satellite Imagery using ML models
M.Tech. (Data Science), Mehta Family School of DS & AI, IIT Roorkee

6. Daksh Yadav 2023–2025
“Disentangling Stock Return and Social Media Sentiment Dynamics: A Multilayer Network Analysis of US Firms”
M.Tech. (Applied Mathematics and Scientific Computing), IIT Roorkee
Co-Supervisor: Prof. Jaydev Dabas

7. Amit Kumar Singh 2024–2026
Investigating the impact of sentiments on stock market using digital proxies
M.Tech. (Data Science), Mehta Family School of DS & AI, IIT Roorkee

8. Ruhul Amin Laskar 2024–2026
Comparative Analysis of News vs Social Media sentiments in forecasting Stock Markets
M.Tech. (Data Science), Mehta Family School of DS & AI, IIT Roorkee

9. Vishal Ramkrishna Patil 2024–2026
Advanced Financial Intelligence via Domain-Adapted Large Language Models and Multi-Source Aggregation
M.Tech. (Data Science), Mehta Family School of DS & AI, IIT Roorkee

Reviewing Services

Journal of Asian Economics; Energy Systems; Heliyon; Discover Sustainability; Quality and Quantity; Engineering Applications of Artificial Intelligence and others

Professional Memberships

Western Economic Association International (WEAI); The Indian Econometric Society (TIES)

Languages

English	Full native and bilingual fluency (completed entire education in this language)
German (Level I)	Limited working proficiency
Hindi	Full native and bilingual fluency
Punjabi	Full native and bilingual fluency

References

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